

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Scotch-Brite™ 7448, Part Number 3823258, or equivalent

Additional Service Items

- No additional service items required.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the intake manifold air pressure/ temperature sensor from the engine harness. Refer to Procedure 019-159 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-159.html>)
- Disconnect the wire harness from the fuel rail pressure sensor.
- Disconnect the wire harness from the cylinder head pass-through connector.
- Disconnect the wire harness attachments from the air intake manifold.
- Remove the charge air piping connection to the air intake manifold. See equipment manufacturer service information.
- Remove the air intake connection adaptor from the air intake manifold, if applicable. Refer to Procedure 010-131 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-131.html>)
- Remove the intake throttle actuator from the air intake manifold, if applicable. Refer to Procedure 010-140 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-140.html>)
- Remove the intake air shutoff valve from the air intake manifold, if applicable. Refer to Procedure 010-143 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-143-tr.html>)

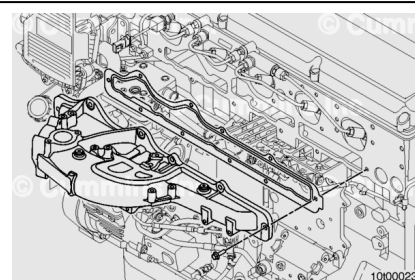
- Remove the exhaust gas recirculation (EGR) crossover tube. Refer to Procedure 011-070 in Section 11. (/qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html)
- Remove the fuel rail pressure relief valve fuel drain line from the engine. Refer to Procedure 006-013 in Section 6. (/qs3/pubsys2/xml/en/procedures/377/377-006-013.html)
- Remove the ether injection hardware from the air intake manifold, if applicable. See equipment manufacturer service information.
- Remove the oil dipstick attachment hardware from the air intake manifold. See equipment manufacturer service information.

Remove

Remove the capscrews that hold the air intake manifold to the cylinder head.

Remove and discard the gasket.

Cover the cylinder head intake ports to prevent debris from entering the engine while the air intake manifold is removed.



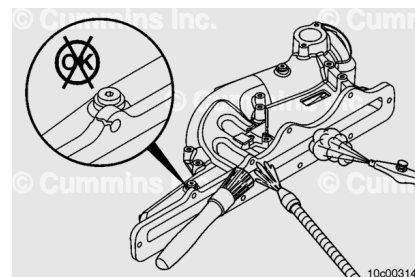
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : Before cleaning the inside of the air intake manifold, make sure to remove the air pressure/temperature sensor.

Clean the intake manifold gasket surface.

Clean the air intake manifold. Saturate the inside of the air intake with safety solvent.

Mineral based solvents can be used if the expected results with the safety solvent are **not** obtained.

Clean the debris with a non-abrasive brush.

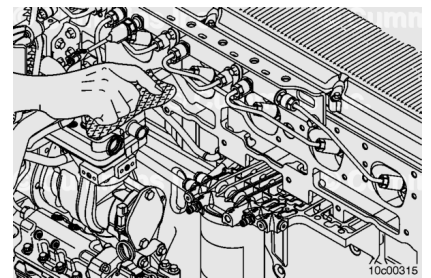
Dry the air intake manifold with compressed air.

Inspect the air intake manifold for cracks or other damage.

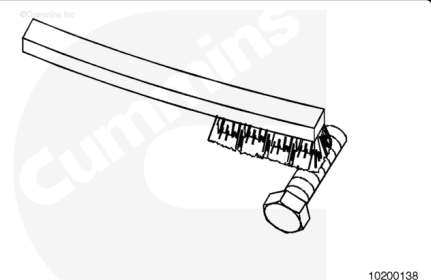
Replace the air intake manifold if damage is found

When inspecting the air intake manifold for oil or debris from the air system malfunction, also inspect the intake ports in the cylinder head for oil and debris.

Some soot buildup inside the cylinder head is normal and will **not** affect engine performance. Clean the air intake to cylinder head surface with Scotch-Brite™ 7448, Part Number 3823258, or equivalent, abrasive pad or a gasket scraper. Do **not** use power tools combined with an abrasive pad to clean the gasket surfaces.

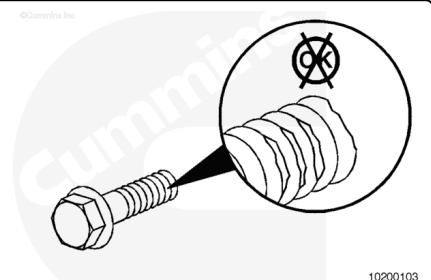


Clean the intake manifold mounting capscrews.



Inspect the air intake manifold mounting capscrews for stripped threads and other damage.

Replace the mounting capscrews if damage is found.



Install

If previously taped off, remove the tape from the cylinder head prior to installing the intake manifold.

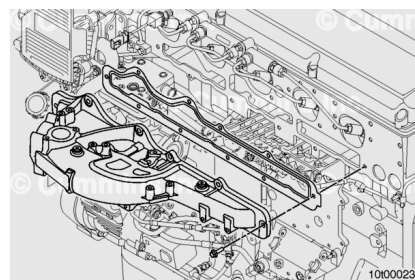
Install two guide pins in the intake manifold capscrew mounting holes in the cylinder head in Number 6 and Number 8 holes before installing a new gasket.

Install the air intake manifold gasket over the guide pins.

Install the air intake manifold over the guide pins.

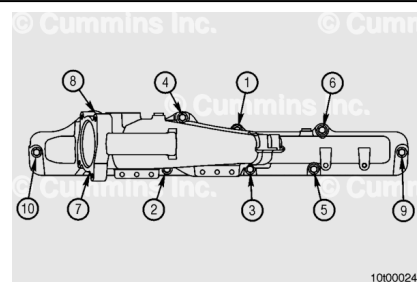
Install the remaining capscrews. Hand tighten the capscrews.

Remove the two guide pins and finish installing the remaining two capscrews hand-tight.



Tighten all the capscrews in the sequence illustrated.

Torque Value: 61 N·m [45 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fuel rail pressure relief valve fuel drain line to the engine. Refer to Procedure 006-013 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-013.html>)
- If removed for cleaning, install and connect the intake manifold air pressure/temperature sensor to the engine harness. Refer to Procedure 019-159 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-159.html>)
- Connect the wire harness to the cylinder head pass-through connector
- Connect the wire harness to the fuel rail pressure sensor.
- Secure the wire harness to the top of the air intake manifold.
- Install the air intake connection adaptor to the air intake manifold, if applicable. Refer to Procedure 010-131 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-131.html>)

- Install the intake throttle actuator to the air intake manifold, if applicable. Refer to Procedure 010-140 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-140.html>)
- Install the intake air shutoff valve to the air intake manifold, if applicable. Refer to Procedure 010-143 in Section 10. (</qs3/pubsys2/xml/en/procedures/377/377-010-143-tr.html>)
- Install the charge air piping. See equipment manufacturer service information.
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Install the ether injection hardware on the air intake manifold, if removed. See equipment manufacturer service information.
- Install the oil dipstick attachment hardware to the air intake manifold. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.